M3 · CAD Module 3, engineering graphics, orthographic projections and building drawing

Engineering graphics: projections, solids, isometric & CAD

BIS SP 46 line conventions and dimensioning systems, first-angle vs. third-angle orthographic projections, solids and isometric views, and NBC SP 7 residential floor plan blueprints.

○ Mark as done ○ Highlight Quiz this module · FAT PYQs

All syllabus sections & lecture slide diagrams included · Ticks itself when every tab is read

BIS Lines & Scales ✓ Dimensioning Rules 1st vs 3rd Angle Projections of Solids Isometric Scale NBC Floor Plans

Official Syllabus Coverage: BIS Lines, Lettering & Dimensioning • 1st/3rd Angle • Solids, Isometric & Perspective • IS SP7 Floor Plan & CAD

Primary Sources: VIT Module 3 Graphics Workbook • SMEC VIT Module 3 & 4 Notes

M3T1 Module 3, topic 1

1. Importance of Engineering Graphics & BIS Standards (IS 10711 / IS SP 46)

Concepts & Working ✓ Key Equations Slide Diagrams Solved PYQ

Overview & context

Why do we call Engineering Drawing the **"Universal Graphic Language of Engineers"**? Spoken languages fail when describing a complex 3D turbine blade or building layout, whereas a drawing prepared to **Bureau of Indian Standards (BIS — IS 10711 / SP 46)** communicates exact shape, size, tolerances, and finish to any machinist anywhere in the world (VIT Module 3 Reference Material).

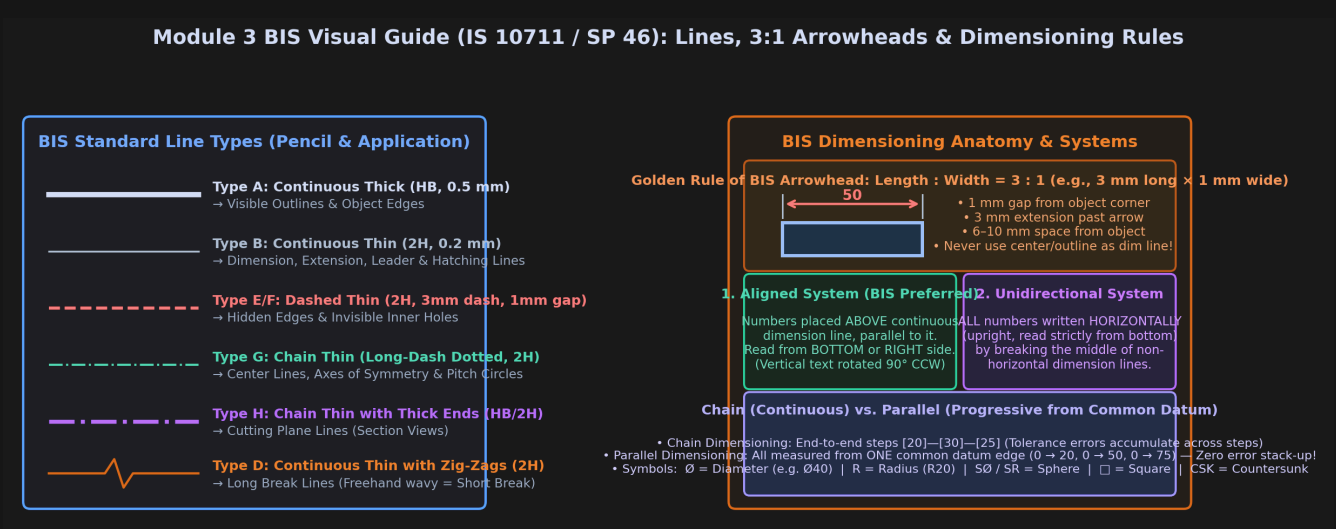


Figure 3.1: BIS Line Types (HB vs 2H), the 3:1 Arrowhead Rule, Aligned vs. Unidirectional, and Chain vs. Parallel Dimensioning.

BIS Standard Topic	Exact Numbers to Memorize	Visual Rule / Pencil Grade	FAT Exam Trap & Shortcut
Drawing Sheet Sizes (IS 10711)	**A0:** 841 × 1189 mm (Area = 1 m^2) **A1:** 594 × 841 mm | **A2:** 420 × 594 mm **A3:** 297 × 420 mm | **A4:** 210 × 297 mm	Aspect Ratio of every sheet is **1 : √2 (1 : 1.414)**. Fold long side in half to get the next size!	**Title Block:** Strictly **185 mm × 65 mm** placed at the **bottom-right corner** of the sheet.
Scales (Representative Fraction RF)	**RF = (Length in Drawing) / (Actual Object Length)** (in same units!)	• **Full Size:** 1 : 1 (RF = 1) • **Reducing:** 1:2, 1:5, 1:100 (Buildings) • **Enlarging:** 2:1, 5:1, 10:1 (Micro gears)	Never write units on RF! An RF of 1:50 means 1 cm on paper = 50 cm in reality.
Types of Lines (IS SP 46)	**Thick (HB, 0.5 mm)** vs. **Thin (2H, 0.2 mm)** (Thickness ratio = 2.5 : 1)	• **Cont. Thick (A):** Visible outlines • **Cont. Thin (B):** Dimension, Extension, Leader, Hatching (45°) • **Dashed Thin (E/F):** Hidden edges • **Chain Thin (G):** Center/Symmetry lines	**Precedence of Overlapping Lines:** 1. Visible Outline (A) wins first! 2. Hidden Dashed Line (E) second! 3. Center Line (G) third! 4. Projection/Extension line last.
BIS Lettering (IS 9609)	**Main Title:** 6–8 mm (or 7 mm) **Subtitles:** 3.5–5 mm **Dimensions / Notes:** 3.5 mm **Drawing Number:** 10–12 mm	**Single-Stroke Gothic Capital Letters:** "Single-stroke" means uniform line thickness obtained in one stroke of pencil (NOT written without lifting pencil!).	Height-to-Width ratio = **7:5 or 6:5**. Vertical (Type B) or Inclined at **75° to horizontal** (15° right slant).
BIS Dimensioning Rules	**Arrowhead Ratio:** Length : Width = **3 : 1** (closed and filled). Extension line gap = 1 mm; extends **3 mm beyond arrow**.	• **Aligned System (BIS Default):** Numbers placed ABOVE continuous line, read from Bottom or Right. • **Unidirectional:** All numbers upright horizontal; break non-horizontal line.	• **Chain Dimensioning:** End-to-end steps. • **Parallel (Datum):** All from one reference edge (zero tolerance stack-up!). • **Leader Line:** Never < 30° (use 45°/60°).

M3T2 Module 3, topic 2

2. Orthographic Projections: 1st Angle (BIS) vs. 3rd Angle (USA)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context

An **Orthographic Projection** ("Ortho" = 90° right angle) flattens a 3D object onto perpendicular 2D screens—the **Vertical Plane (VP, for Front View / Elevation)** and **Horizontal Plane (HP, for Top View / Plan)**—using parallel rays of sight perpendicular to the plane (SMEC VIT Module 3 & 4 Full Notes). After projecting, the Horizontal Plane is ALWAYS rotated **90° clockwise downward** so the sheet lies flat!

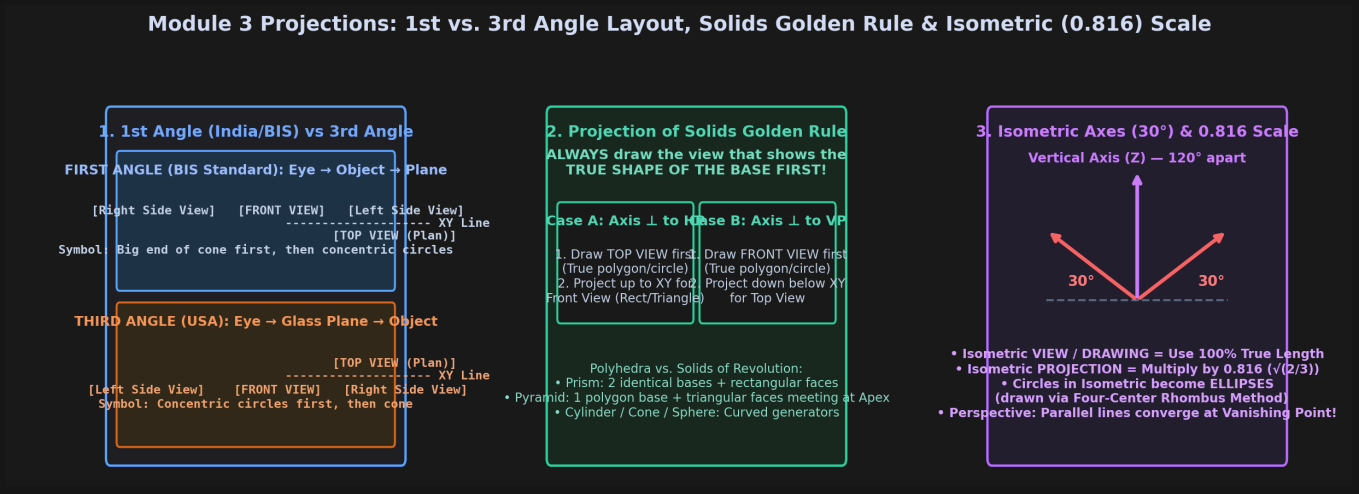


Figure 3.2: 1st Angle vs. 3rd Angle Projection layouts, Projection of Solids Golden Rule, and Isometric (0.816) vs. Perspective Projection.

Real-world intuition

The "Cinema Projector vs. Glass Fishbowl" Analogy for 1st vs. 3rd Angle

First Angle Projection

Think of a **Cinema Screen behind the actor**: **[Observer's Eye] → [Object] → [Opaque Projection Plane]**. Because the object sits between you and the wall, its shadow lands on the *opposite* side! When HP rotates 90° down: **Top View is BELOW Front View**, and **Left Side View is drawn on the RIGHT of Front View**.

Third Angle Projection

Think of tracing an object inside a **Transparent Glass Box**: **[Observer's Eye] → [Transparent Plane] → [Object]**. Because the glass screen is right in front of each face, every view stays on its *own* side: **Top View is ABOVE Front View**, and **Left Side View is on the LEFT of Front View**.

Why do we NEVER use 2nd or 4th Angle Projection?

Because when the Horizontal Plane (HP) is rotated 90° clockwise in Quadrant 2 (both above XY) or Quadrant 4 (both below XY), the **Front View and Top View overlap on top of each other**, making the drawing unreadable!

Feature	First Angle Projection (BIS / ISO Standard)	Third Angle Projection (USA / ANSI Standard)
Object Position & Order	Placed in **1st Quadrant** (Above HP, In front of VP). Order: **Observer → Object → Plane**.	Placed in **3rd Quadrant** (Below HP, Behind VP). Order: **Observer → Transparent Plane → Object**.
Nature of Plane	Planes are assumed **Non-Transparent (Opaque)**.	Planes are assumed **Transparent (Glass Box)**.
Position of Views Around Front View	• **Top View (Plan):** Strictly **BELOW** XY line (below FV). • **Left Side View:** Drawn on **RIGHT** of Front View.	• **Top View (Plan):** Strictly **ABOVE** XY line (above FV). • **Left Side View:** Drawn on **LEFT** of Front View.

Feature	First Angle Projection (BIS / ISO Standard)	Third Angle Projection (USA / ANSI Standard)
	• **Right Side View:** Drawn on **LEFT** of Front View.	• **Right Side View:** Drawn on **RIGHT** of Front View.
BIS Symbol (Truncated Cone)	Trapezoid (side view of cone) on left; two concentric circles on right (looking at small end, projected past it).	Two concentric circles on left; trapezoid on right (cone points toward the circles).

M3T3 Module 3, topic 3

3. Projection of Solids, Isometric Projection (0.816) & Perspective

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context

A **3D Solid** has three dimensions (length, breadth, height/axis) and requires at least two orthographic views (Front View + Top View) to define completely (VIT Module 3 Workbook).

Polyhedra (Flat Faces)

(1) **Regular Polyhedra (Platonic Solids):** All faces identical regular polygons—*Tetrahedron* (4 triangles), *Cube/Hexahedron* (6 squares), *Octahedron* (8 triangles), *Dodecahedron* (12 pentagons), *Icosahedron* (20 triangles). (2) **Prisms:** Two identical parallel polygon bases joined by **rectangular lateral faces** (Triangular, Square, Pentagonal, Hexagonal prism). (3) **Pyramids:** One polygon base and **isosceles triangular slant faces** meeting at a single top point called the **Apex / Vertex**.

Solids of Revolution (Curved Surfaces)

Cylinder (rectangle revolved around one side), **Cone** (right triangle revolved around perpendicular side), and **Sphere** (semi-circle revolved around diameter—projects as a circle of true diameter in EVERY view!).

The Golden Exam Rule for Simple Position of Solids

Always draw the view that shows the **True Shape of the Base (Polygon or Circle) FIRST**! If the axis is **perpendicular to HP (⊥ HP)**, the base rests parallel to HP → **draw Top View (circle/polygon) first**, then project up for the Front View. If the axis is **perpendicular to VP (⊥ VP)**, the base faces VP → **draw Front View first**, then project down for the Top View!

Exam memory hook

Isometric Projection (0.816 Scale) vs. Isometric View (100% True Length) vs. Perspective

Isometric Axes

Three axes (X, Y, Z) equally inclined to the plane of projection, separated by **120°** from each other. On paper, vertical edges stay **vertical (90°)**, while left and right horizontal base edges recede at **30° to the horizontal**!

Isometric PROJECTION vs	Because edges are tilted away from the viewer, their true length foreshortens by **cos 45° / cos 30° = √(2/3) ≈ 0.8165 (81.6%)**. An **Isometric Projection** is drawn using this reduced **Isometric Scale (0.816 × True Length)**. An **Isometric View (or Isometric Drawing)** is drawn directly using **100% True Lengths** (no scaling needed, so it looks ~22.5% larger).
How to Draw Circles & Solids in Isometric	Enclose polygons in a rectangular box (**Box Method**); draw non-isometric slant lines by locating their endpoint coordinates (**Offset Method**). A circle on an isometric face becomes an **Ellipse** drawn inside a 30° rhombus using the **Four-Center Method**!
Perspective Projection (Central Projection)	Projectors are NOT parallel—they **converge at a single Station Point (Observer's Eye)** and parallel receding edges meet at a **Vanishing Point (VP)** on the horizon. Gives a realistic camera/architectural view, but **does not preserve true dimensions or parallel lines** (drawn via Visual Ray or Vanishing Point method).

M3T4 Module 3, topic 4

4. Freehand Sketching, IS SP7 Residential Floor Plan & Introduction to CAD

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	In real engineering brainstorms and VIT FAT exams, you must be able to sketch machine parts freehand (holding the pencil **30–40 mm from the tip**, pulling light 2H construction lines toward your body, drawing circles via the **Centerline + Enclosing Square 8-Point Method**, and darkening final outlines with HB) AND draw a complete **Single-Bedded (1BHK) Residential Floor Plan as per IS SP7 (National Building Code of India)** (SMEC VIT Module 3 & 4 Full Notes).

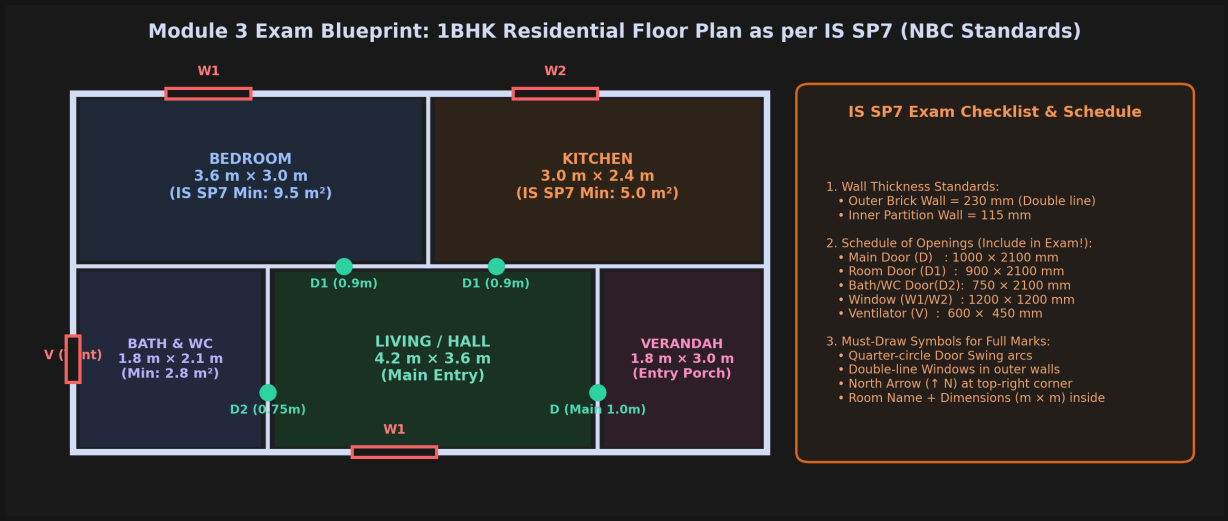


Figure 3.3: Exam-ready 1BHK Residential Floor Plan as per IS SP7 with NBC minimum room dimensions, door swings, windows, and Schedule of Openings.

IS SP7 / CAD Topic	Standard NBC / IS SP7 Specifications	How to Score 10/10 in the Exam
What is a Floor Plan?	A horizontal section view cut at **Window Sill Level (~1.0 to 1.2 m above floor)** looking straight down after removing the top half of the building.	Always draw **double lines for wall thickness**: Outer brick walls = **230 mm (9 inches)**; Inner partition walls = **115 mm (4.5 inches)**.
IS SP7 Minimum Room Sizes (NBC)	• **Living / Drawing Room:** 4.2 m × 3.6 m (Min area ≥ 9.5 m^2, width ≥ 2.4 m) • **Bedroom:** 3.6 m × 3.0 m (Min area ≥ 9.5 m^2) • **Kitchen:** 3.0 m × 2.4 m (Min area ≥ 5.0 m^2, width ≥ 1.8 m)	• **Combined Bath & WC:** 1.8 m × 2.1 m (Min ≥ 2.8 m^2) • **Verandah:** 1.8 m × 3.0 m. Write **Room Name + Dimensions (m × m)** inside every room box and add a **North Arrow (↑ N)**!
Schedule of Openings (IS SP7)	• **Main Door (D):** 1000 × 2100 mm • **Room Door (D1):** 900 × 2100 mm • **Bath/WC Door (D2):** 750 × 2100 mm	• **Window (W1/W2):** 1200 × 1200 mm • **Ventilator (V):** 600 × 450 mm. Always draw quarter-circle door swings opening against a wall!
Introduction to CAD (Computer-Aided Design)	Software tools (AutoCAD, SolidWorks, CATIA, Fusion 360, Creo) used to create, modify, analyze, and optimize 2D drawings and 3D solid/surface models.	**7 Core CAD Applications:** (1) 2D Drafting, (2) 3D Wireframe/Surface/Solid Modeling, (3) Assembly & Clash Detection, (4) FEA/CFD Simulation, (5) Optimization, (6) PDM Collaboration, (7) Direct CAD/CAM & .STL export.

M3T5 Module 3, topic 5

5. Module 3 FAT PYQ Self-Assessment & Complete Model Answers

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	Practice these high-probability 10-mark VIT Module 3 questions (VIT SMEC BAEEE101 Question Bank & CAT-2 E1 Paper).

PYQ 1 [10 Marks — VIT CAT-2 / FAT]: First Angle vs. Third Angle Projection & BIS Line Conventions

Overview & context	**Question:** (a) Differentiate between First Angle Projection and Third Angle Projection with neat BIS standard symbols and explain why Second and Fourth

	Angle projections are not used in engineering practice. (b) Sketch and state the BIS application of any five standard line types as per IS SP 46.
Overview & context	**Model Answer:**
Part (a) 1st vs. 3rd Angle (5 Marks)	Present the 4-point comparison table from Section 2 (Quadrant 1 vs. 3, Opaque vs. Transparent plane, Top View below XY vs. above XY, Left Side View on Right vs. on Left) and draw the **Truncated Cone symbols** (Figure 3.2). Explicitly state: *"Second and Fourth angle projections are never used because when the Horizontal Plane (HP) is rotated 90° clockwise to align with the Vertical Plane (VP), both HP and VP overlap on the same side of the XY reference line (both above XY in 2nd quadrant; both below XY in 4th quadrant), causing the Front View and Top View to superimpose and become unreadable."*
Part (b) 5 BIS Line Types (5 Marks)	Sketch as shown in Figure 3.1: (1) **Continuous Thick (Type A, HB 0.5 mm):** Visible outlines and edges. (2) **Continuous Thin (Type B, 2H 0.2 mm):** Dimension lines, extension lines, leader lines, and 45° section hatching. (3) **Dashed Thin (Type E/F, 2H):** Hidden/invisible edges and inner contours. (4) **Chain Thin Long-Dashed Dotted (Type G, 2H):** Center lines, axes of symmetry, and pitch circles. (5) **Chain Thin with Thick Ends (Type H):** Cutting-plane lines.
PYQ 2 [10 Marks — VIT Graphics Workbook / FAT]: Projection of a Hexagonal Prism & Isometric Cylinder/Cone	
Overview & context	**Question:** (a) A regular hexagonal prism of base side 25 mm and axis height 60 mm rests on its base on the Horizontal Plane (HP) with one of its base edges parallel to the Vertical Plane (VP). Write the step-by-step construction and draw its Front View and Top View in First Angle Projection. (b) Distinguish between Isometric Projection and Isometric View, and explain the Four-Center Method used to draw the isometric view of a cylinder or cone of base diameter 50 mm and height 70 mm.
Overview & context	**Model Answer:**
Part (a) Hexagonal Prism Construction (5 Marks)	(1) Draw horizontal reference line **XY**. Since the axis (60 mm) is perpendicular to HP, the base rests on HP—so by the Golden Rule, **draw the Top View first below XY**! (2) Below XY, construct a regular hexagon *abcdef* (top face) and *(1)(2)(3)(4)(5)(6)* (bottom face) of side **25 mm** with two horizontal edges parallel to XY. (3) Project vertical thin construction lines (2H) upward from all 6 corners to XY. (4) Above XY, draw the base on XY and extend vertical edges **60 mm** upward to form the rectangular Front View *a'b'c'd' – 1'2'3'4'*, darkening the outer boundary and visible vertical edges with an HB pencil and marking dimensions (25 mm and 60 mm) per the Aligned System.
Part (b) Isometric Scale & Four-Center Ellipse Method (5 Marks)	(1) **Isometric Projection** foreshortens all dimensions along the 30° isometric axes by **Isometric Scale = √(2/3) = 0.816 × True Length**, whereas an **Isometric View/Drawing** uses **100% True Length** directly. (2) **Four-Center Method for Cylinder/Cone Base (D = 50 mm):** Draw an isometric rhombus ABCD of side 50 mm with edges inclined at 30° to horizontal. Mark the midpoints (M_1, M_2, M_3, M_4) of the 4 sides. Join the two obtuse corners (B and D) to the midpoints of the opposite sides. The intersections of these lines give 4 arc centers (B, D, C_1, C_2) to draw 4 smooth tangent circular arcs forming the isometric ellipse! For a **Cone**, erect a vertical axis of 70 mm from the rhombus center to the Apex O and draw

	two tangent generators to the base ellipse; for a **Cylinder**, repeat the ellipse 70 mm higher and join extreme edges with vertical tangents.
PYQ 3 [10 Marks — VIT FAT High-Weightage Question]: Single-Bedded Residential Floor Plan as per IS SP7	
Overview & context	**Question:** Draw a neat dimensioned double-line floor plan of a single-bedded (1BHK) residential building as per IS SP7 (National Building Code) showing Living Room, Bedroom, Kitchen, Combined Bath/WC, and Verandah, complete with standard architectural symbols and a Schedule of Openings.
Overview & context	**Model Answer Blueprint (Reproduce Figure 3.3 in Exam for 10/10):**
Step 1 — Outer & Inner Walls (3 Marks)	Draw the outer rectangular footprint using double lines (230 mm outer wall thickness) and divide into 5 functional zones using 115 mm inner partition walls: **Verandah** (1.8 m × 3.0 m) leading via Main Door **D** into **Living/Drawing Room** (4.2 m × 3.6 m), connected via internal doors **D1** to **Bedroom** (3.6 m × 3.0 m) and **Kitchen** (3.0 m × 2.4 m), plus **Combined Bath & WC** (1.8 m × 2.1 m) via door **D2**.
Step 2 — Architectural Symbols (4 Marks)	Draw quarter-circle **door swing arcs** opening against the nearest wall (never blocking room center), double-line **Window symbols (W1, W2)** centered on external walls for cross-ventilation, a **Ventilator (V)** on the Bath/WC outer wall, and a **North Arrow (↑ N)** at the top-right corner.
Step 3 — Schedule of Openings Table (3 Marks)	Draw the 5-row Schedule table beside the plan: **D (Main Door):** 1000 × 2100 mm; **D1 (Bedroom/Kitchen Door):** 900 × 2100 mm; **D2 (Bath/WC Door):** 750 × 2100 mm; **W1/W2 (Window):** 1200 × 1200 mm; **V (Ventilator):** 600 × 450 mm.
PYQ 4 [10 Marks — VIT FAT Theory]: Isometric vs. Perspective Projection & 7 Core Applications of CAD	
Overview & context	**Question:** (a) Compare Isometric Projection and Perspective Projection across four geometric parameters. (b) Explain how Computer-Aided Design (CAD) transforms the modern engineering design workflow across seven key stages.
Overview & context	**Model Answer:**
Part (a) Isometric vs. Perspective (4 Marks)	(1) **Projector Rays:** Isometric uses **parallel projectors** perpendicular to the projection plane (axonometric); Perspective uses **converging projectors** meeting at a single Station Point (Observer's Eye). (2) **Parallel Edges:** In Isometric, parallel edges on the object remain strictly parallel on paper (at 30°); in Perspective, parallel receding lines converge at a **Vanishing Point (VP)** on the horizon. (3) **Scale & Measurement:** Isometric preserves uniform scale (100% in view or 0.816 in projection) so dimensions can be measured directly; Perspective distorts scale with depth so distant edges shrink. (4) **Application:** Isometric is used for machine parts and piping catalogues; Perspective is used for architectural exteriors and realistic human-eye visualization.
Part (b) 7 Applications of CAD (6 Marks)	(1) **2D Parametric Drafting & Detailing:** Automated BIS dimensioning, hatching, and BOM tables. (2) **3D Geometric Modeling:** Wireframe, Surface, and Solid modeling (CSG & B-Rep). (3) **Digital Assembly & Kinematics:** Testing part mating and collision/interference before manufacturing. (4) **Engineering Analysis (FEA / CFD):** Simulating structural stress, thermal flow, and vibration. (5) **Design Optimization:** Generative weight reduction while maintaining safety factor. (6) **Collaborative Cloud PDM/PLM:** Multi-team version control. (7) **Direct CAD/CAM & Additive Export:** Generating CNC G-code and 3D-printable **.STL** triangle meshes.